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Primary tumor location impacts breast cancer survival

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Abstract

BACKGROUND: The prognostic significance of tumor location in breast cancer remains unclear. To better understand this relationship, we evaluated the Department of Defense tumor registry.

METHODS: Patients with infiltrating ductal adenocarcinoma or lobular carcinoma over a 10-year period were identified and analyzed.

RESULTS: Of the 13,984 tumors, 7,871 (58%) originated from the upper-outer quadrant or axillary tail, whereas the remainder were found at the nipple complex (9%), upper-inner quadrant (14%), lower-inner quadrant (9%), and lower-outer quadrant (10%). Univariate analysis of cancer-specific survival revealed a significant difference based on location of the primary breast cancer. Upper-outer quadrant lesions were associated with an independent contribution toward a survival benefit.

CONCLUSIONS: Upper-outer quadrant breast cancers have a more favorable survival advantage when compared with tumors in other locations. Factors that negatively impacted survival included high-grade tumors, advanced stage, and race.

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Although nodal status, tumor size, and hormone-receptor status, among other factors, clearly have been linked to breast cancer survival, the impact of location on survival has been defined less clearly. Recently, investigators have concluded that breast cancers located in medial, central, or lower quadrants may be associated with adverse survival and that these patients may benefit from additional therapies.^{1–6} Their premise, that lower- and medial-quadrant breast tumors are anatomically closer and more likely to metastasize to the often unevaluated internal mammary

lymph nodes, remains a matter of contention. If true, the radiographic, surgical, and medical treatments for breast cancer must be re-evaluated. The purpose of our study was to identify the prognostic significance of tumor location in female breast cancer patients.

Methods

In this retrospective cohort review, the prospectively maintained Department of Defense tumor registry was analyzed to identify female patients with breast cancer with ductal and/or lobular variants. Male patients, those with other histologic variants, and in situ disease were excluded from this analysis. In addition, to facilitate analysis, multifocal disease, cancer involving more than one quadrant, and cancer without specific locations were excluded. This study was approved by the local institutional review board.

The Department of Defense tumor registry encompasses all military facilities from the United States Army, Air

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Force, and Navy, with more than 100 institutions from the United States, Germany, and Japan. This unique health care system is designed to deliver equitable health care for active-duty personnel, their family members, and eligible retirees regardless of rank, income, sex, and social status. Once eligible, all health care is free of charge and includes routine screening, physician appointments, and, if appropriate, percutaneous biopsy, surgery, medication, and systemic therapy. Beneficiaries who were treated at civilian community institutions because of a lack of local military installation capabilities were not included in this database.

Characteristic variables from the tumor registry included patient age at presentation, race, stage of disease, location of primary lesion, and grade of tumor. For location, the breast was divided into 5 quadrants: lower-inner, lower-outer, upper-inner, upper-outer and axillary tail, and central including the nipple and areola complex.

The primary outcome variable was cancer-specific survival. Univariate statistics were used to compare patients based on location of the lesion. Kaplan-Meier analysis was used to describe the impact of location on survival, and Cox regression analysis was used to determine the impact of confounding variables.

Results

From January 1990 until December 2005 there were 13,984 out of 26,121 (54%) patients who met the inclusion criteria and comprised our patient cohort. The majority of patients who were excluded from the analysis did not have an accurate record of location or had disease involving more than one quadrant. The demographics for our population are listed in Table 1. The mean patient age was 56 years (± 15 y), and the population for analysis was predominately white. The majority of the breast cancers were located in the upper-outer quadrant (57%); the remaining locations were distributed equally with the exception of very few cancers located in the axillary tail. With respect to grade, the cancers were distributed equally between grades 1 to 3, but there were few grade 4 tumors. The same trend was true for stage of disease. The majority of cancers were stage I (48%) and stage II (40%), whereas only 9% were stage III and 2% were stage IV.

The relationship of patient characteristics then was evaluated with respect to breast cancer location (Table 2). The cancers located in the upper-outer quadrant, lower-outer quadrant, and the axillary tail presented at significantly younger ages compared with cancers located at the nipple or inner quadrants. This difference, although statistically significant, only equated to an absolute difference of 2 to 3 years in presentation. Statistical significance ($P < .05$) was also reached for grade, race, and stage of disease based on breast cancer location. By examining Table 2, there were subtle differences within these variables, but the percentage differences were extremely small when comparing grade and race.

Table 1 Population demographic characteristics

Variable	Patients
Mean age, y ($\pm$ SD)	56 (15)
Race, n (%)	
White	15,906 (75)
Black	2,691 (13)
Asian	1,782 (8)
Hispanic	796 (4)
Location, n (%)	
Nipple complex	1,304 (9)
Upper-inner quadrant	2,007 (14)
Lower-inner quadrant	1,222 (9)
Upper-outer quadrant	7,970 (57)
Lower-outer quadrant	1,380 (10)
Axillary tail	101 (1)
Grade, n (%)	
1: Well-differentiated	3,620 (23)
2: Moderately differentiated	6,555 (30)
3: Poorly differentiated	5,600 (35)
4: Undifferentiated	227 (1)
AJCC stage, n (%)	
I	7,842 (48)
II	6,530 (40)
III	1,424 (9)
IV	391 (2)
Mean follow-up period, y ($\pm$ SD)	6 (4)

Univariate analysis of cancer-specific survival revealed a significant difference based on location of the primary breast cancer (log-rank test, $P < .05$). By using Cox regression analysis, several factors impacted the overall survival, including age, race, grade of tumor, stage of disease, and tumor location (Table 3). Age and black race imparted a negative impact on survival (hazard ratio [HR], 1.035; 95% confidence interval [CI], 1.031–1.039; and HR, 1.35; 95% CI, 1.172–1.554, respectively). The upper-outer quadrant was the only location that impacted survival when adjusting for other patient characteristics (HR, .820; 95% CI, .820–.960; $P < .05$). The trend of this impact was for upper-outer quadrant lesions to impart an independent survival benefit as compared with other locations within the breast. Grade of tumor and stage of disease also were significant in the multivariate regression analysis. Increasing grade contributed to a worsening survival. In addition, increasing stage of disease imparted a negative impact on survival, and this variable also contributed the largest magnitude with a HR of 4.05 for stage III and a HR of 10.7 for stage IV disease.

Comments

In a health care system designed to deliver equitable care, analysis of the female breast cancer cohort over a 10-year period found a more favorable survival advantage for tumors originating in the upper-outer quadrants when compared with tumors in other locations. Similar to other studies, tumors were found most commonly in this upper-

Table 2 Comparison of patient characteristics by breast cancer location

Variable	Nipple complex	Upper-inner quadrant	Lower-inner quadrant	Upper-outer quadrant	Lower-outer quadrant	Axillary tail	P value
Mean age, y ($\pm$ SD)	57 ($\pm$ 13)	56 ($\pm$ 13)	57 ($\pm$ 13)	55 ($\pm$ 13)*	55 ($\pm$ 13)*	53 ($\pm$ 13)*	<.05
Race, %							<.05
White	9	14	9	58	10	1	
Black	9	14	11	53	12	1	
Asian	9	17	8	55	11	0	
Hispanic	10	13	9	57	10	1	
Grade, %							<.05
1: Well-differentiated	9	15	9	57	9	1	
2: Moderately differentiated	9	15	9	56	10	1	
3: Poorly differentiated	8	15	9	58	10	1	
4: Undifferentiated	10	14	7	58	12	0	
AJCC stage, %							<.05
I	8	17	10	55	10	1	
II	9	13	7	60	10	1	
III	13	11	13	55	11	1	
IV	9	15	9	57	11	0	

Univariate comparisons for age performed by using analysis of variance for multiple continuous variables.

Comparisons of categorical variables were performed using the chi-squared test.

*Denotes the Tukey post hoc analysis, and those marked are significantly different than the remaining variables.

outer region and factors that impacted survival negatively included high-grade tumors, advanced stage, and race.

Although the results of our analysis are neither unique nor conclusive, it contributes to the recent growing body of literature that supports re-evaluating the staging of breast

cancer based on location. The premise that certain tumor locations are more prone to metastasize to the internal mammary chain (IMC) may possibly explain survival differences. Results from randomized trials showing no survival advantage in patients undergoing routine IMC dissection ended sampling of this nodal chain as part of surgical staging.^{7,8} However, interest still remains to further elucidate this relationship. For instance, Shahar et al⁵ recently reviewed the MD Anderson experience and found that lower and central breast tumors were more likely to harbor disease in the IMC. They found that 43% of tumors in the lower-inner quadrant drain to the IMC compared with 30% and 15% for the lower-outer and other quadrants, respectively. Similarly, Sarp et al,⁴ in a review of the Geneva Cancer registry, found that tumors located in the lower-inner quadrants were independently associated with worse prognosis for stage I breast cancers. In addition, in an analysis of the Surveillance, Epidemiology, and End-Results registries, Gaffney et al⁹ found that tumors located medially adversely impacted overall survival. Other investigators have asserted that for patients with medial tumors, a new staging procedure should be devised to sample internal mammary nodes. Colleoni et al,¹ based on the analysis from the International Breast Cancer Study Group, stated that “patients with medial presentation represent a cohort of whom studies of more focused staging procedures are required to assess internal mammary node metastases.”

In light of all of this evidence, why then is this still a controversial topic and not the standard of care? First, other studies have not uniformly found the same association between tumor location and breast cancer prognosis.^{3,6,10,11} This lack of consensus and disparate outcomes based on tumor locations may be in part owing to lack of standard-

Table 3 Cox regression results for independent predictors of cancer-specific survival

Factor	Breast cancer survival* HR (95% CI)	P value
Age (per year)	1.035 (1.031–1.039)	<.01
Race		
White	1.0†	
Black	1.35 (1.172–1.554)	<.01
Asian	.855 (.707–1.035)	.109
Hispanic	.935 (.714–1.223)	.623
Grade		
1	1.0†	
2	1.37 (1.193–1.508)	<.01
3	1.68 (1.456–1.959)	<.01
AJCC stage		
I	1.0†	
II	1.68 (1.51–1.88)	<.01
III	4.05 (3.48–4.72)	<.01
IV	10.7 (8.62–13.29)	<.01
Location		
Nipple complex	1.0†	
Upper-inner quadrant	.840 (.694–1.018)	.075
Lower-inner quadrant	.888 (.717–1.099)	.273
Upper-outer quadrant	.820 (.820–.960)	<.05
Lower-outer quadrant	.864 (.702–1.063)	.168
Axillary tail	.702 (.382–1.290)	.254

*Breast cancer survival HR is for death.

†Represents reference category.

ized division of the breast. Although some investigators based their conclusions on medial and lateral sections of the breast, others have been more specific. This, in effect, causes a dilution of the data that prohibit a combined analysis of the different series. More importantly, in a study evaluating extended radical mastectomy (radical mastectomy with resection of the IMC), Veronesi et al¹² showed an equal rate of IMC metastases for the inner and central quadrants compared with the outer quadrants. This subgroup analysis was based on a randomized, prospective, international study. The study, with a 10-year follow-up period, failed to show a survival advantage of extended radical mastectomy over radical mastectomy.⁷ This trend continued and, more recently, in a study with more than 30 years of follow-up evaluation of these same patients, no survival advantage with dissection of the IMC was seen.¹³ When combined, these studies suggest that association between tumor location and survival indeed may be unrelated, or at the least offer no clinical significance.

Based on our own data, we cannot conclusively make recommendations that tumor location and survival are the result of metastasis to the IMC. However, given the advances in preoperative tumor detection, it is not unreasonable to critically evaluate more than axillary nodal basins of tumors in the lower-medial quadrants. Certainly, further research may be able to identify higher-risk patients more definitively. Alternatively, survival differences based on location noted in our tumor registry may not be attributable to IMC metastasis but rather a function of overall tumor genetics. Certainly, as outlined, there is a recent body of literature to support IMC metastasis may account for some of these differences; however, other factors certainly are possible and should not be dismissed.

There were several limitations to this study that deserve mention. Although the database contains a large number of patients with survival outcomes, it does not contain the definitive medical and surgical treatments that our patients underwent. More specifically, we do not know if and what type of adjuvant therapy, including hormone therapy, these patients received. Presumably, the standard of care was met based on patient and physician preference; however, this was an assumption and a significant unmeasured variable in our study. Reassuringly, survival based on stage was within the accepted ranges.

In conclusion, breast cancers located in the upper-outer quadrant were associated with a survival advantage (albeit

small when compared with the clinical impact of grade and stage of disease on breast cancer survival). In the analysis of our large data set, traditional factors including advanced age, race, high-grade tumor, advanced stage, and negatively impacted survival.

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